SPHEROID METRICS

Automated Segmentation and Morphological Analysis of Cancer Spheroids in Bright-Field Images

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Project Supervisor
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Key Stakeholders
Dr. Dalia Mahdy

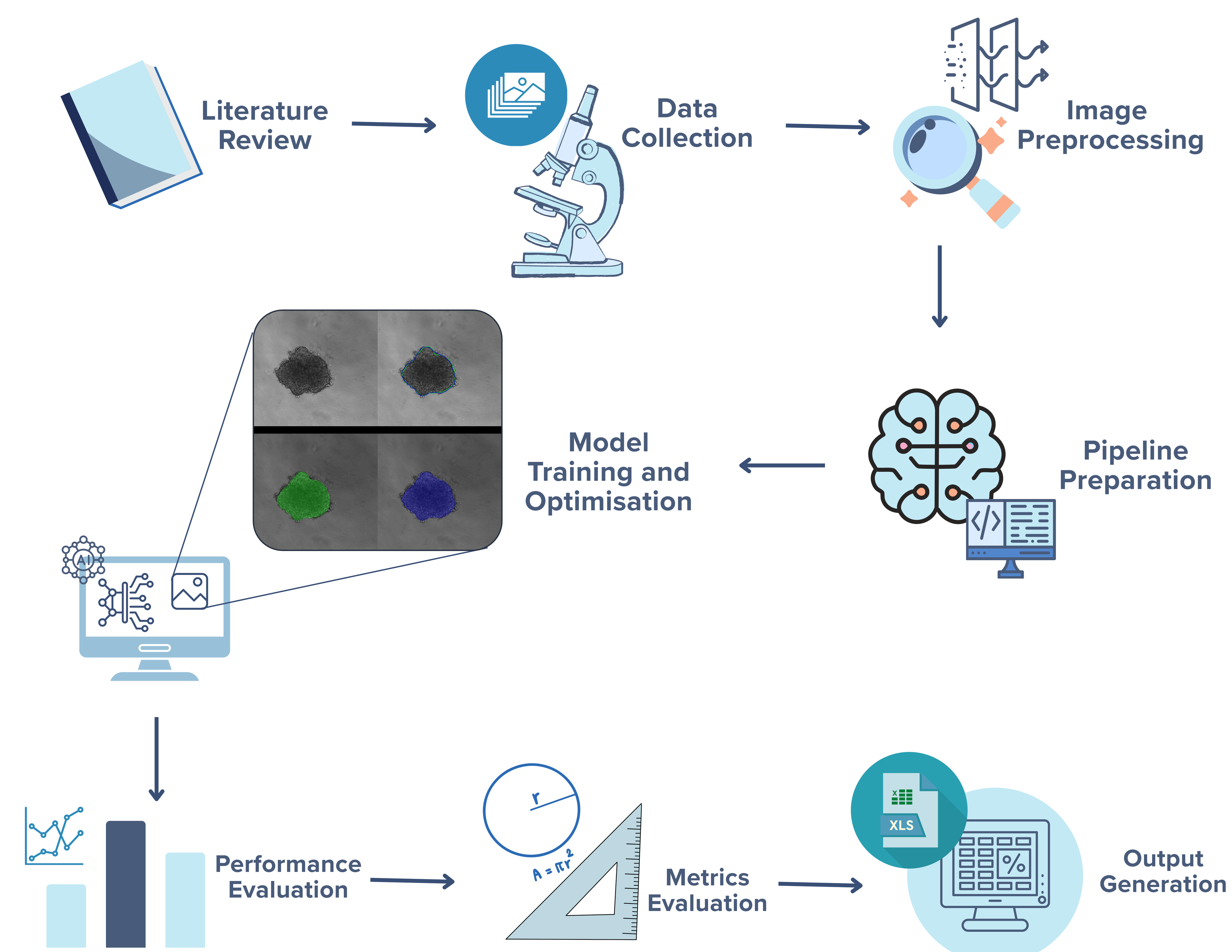
Background

- Pancreatic cancer is highly fatal and hard to treat.
- 3D spheroids offer tissue-like disease models.
- They support studies of tumor growth and drug response.
- Morphological metrics reveal growth and treatment effects.
- Accurate analysis is crucial for reliable results.

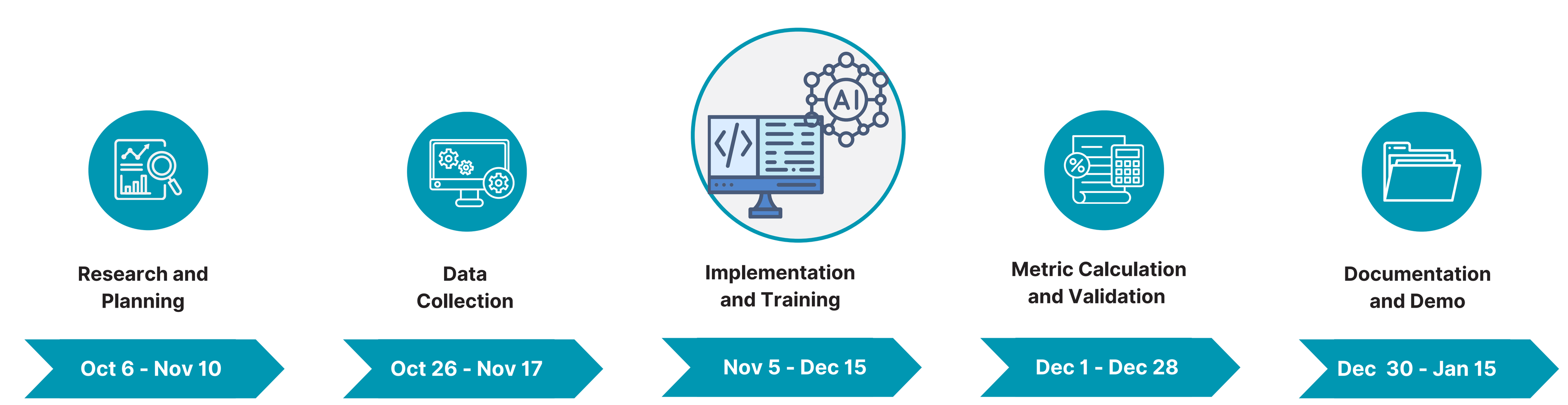
Problem

- Manual spheroid measurement is slow and subjective.
- Inefficient for large datasets or time-point comparisons.
- Automation is needed for faster, consistent analysis.
- Automation enables accurate measurement of morphological metrics such as area, perimeter, roundness, and brightness.

Methodology



Timeline

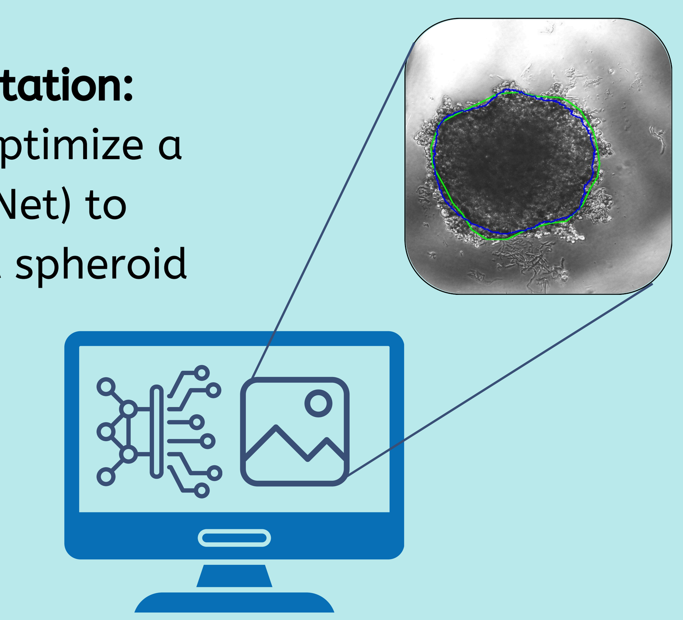


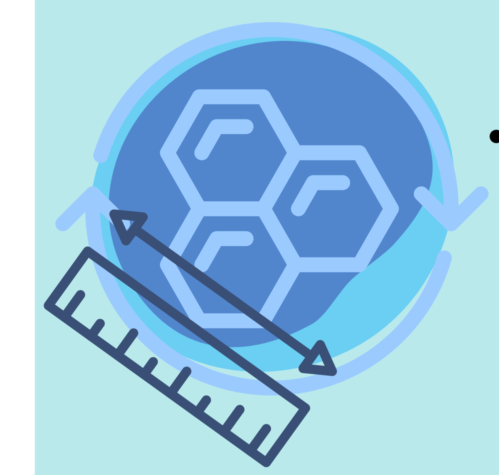
https://github.com/thws-mai/PM25_Efficacy

Objectives

This project aims to develop an automated image-analysis pipeline to:

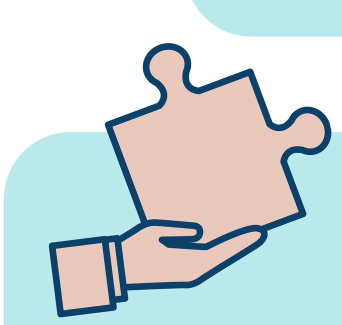
- **Spheroid Segmentation:** Implement and optimize a model (YOLO/U-Net) to accurately detect spheroid boundaries.





- **Feature Extraction:** Calculate key morphological metrics and raw pixel intensity from segmented regions.
- **Feature Comparison:** Perform comparative analysis of metrics to assess growth and drug-induced morphological changes.

Challenges



- Limited data restricts training performance.
- Debris obscures the boundary, making recognition difficult.
- Irregular spheroid shapes and merging over time complicate accurate feature measurement.
- Differences in lighting, focus, or image acquisition between pre-treatment and post-treatment images can affect metric comparison.

Authors

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